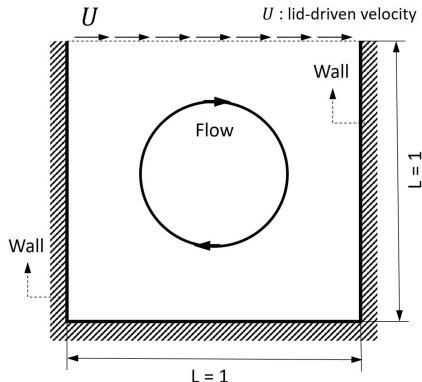


Problem defs & PDEs to solve

Lets try to simulate some basic CFD problem - a square cavity flow - by LBM!. We need to solve following system of equations:

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}$$



LBM algorithm

- Initialize DFs $f_i^{\text{eq}}(\rho_{\text{ini}}, \mathbf{u}_{\text{ini}})$
- Setup τ according to Reynolds number
- Repeat until steady state or desired time is achieved
 1. Compute macroscopic quantities from f_i
 2. Compute equilibrium DF's and post-collision state
 3. Stream f_i^* along lattice links
 4. Apply BC

LBE Equations, when D₂Q₉ model is used, we have 9 such eqns.

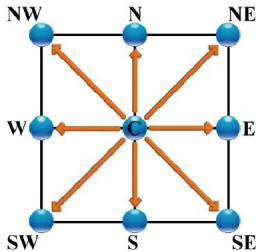
$$f_i(x + c_i, t + 1) = f_i(x, t) + \frac{1}{\tau} [f_i^{\text{eq}}(x, t) - f_i(x, t)]$$

with f_i^{eq} and τ defined by

$$\nu_{lb} = \frac{1}{3} \left(\tau - \frac{1}{2} \right), \quad f_i^{\text{eq}} = t_i \rho(x, t) \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{\mathbf{c}_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right)$$

LBM D2Q9 Model

2D model with 9 discrete velocities



	C	E	N	W	S	NE	NW	SW	SE
$\mathbf{c}_i$	(0,0)	(1,0)	(0,1)	(-1,0)	(0,-1)	(1,1)	(-1,1)	(-1,-1)	(1,-1)
t_i	4/9	1/9	1/9	1/9	1/9	1/36	1/36	1/36	1/36

lattice speed of sound $c_s = \frac{1}{\sqrt{3}}$

Dimless numbers & macro variables

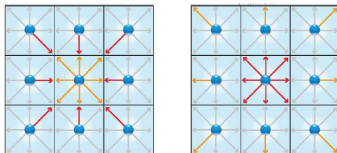
Reynolds and Mach numbers could be used to setup the LBM

$$u_{x,lb} = Ma \sqrt{c_s^2} \quad \nu_{lb} = \frac{u_{x,lb} L_{Y,lb}}{Re} \quad \tau = \nu_{lb} / c_s^2 + .5$$

Macro variables are obtained by

$$p/c_s^2 = \rho = \sum_i f_i(t, \mathbf{x}), \quad \rho \mathbf{u} = \sum_i \mathbf{c}_i f_i$$

Streaming & BCs



Advanced BC's are tricky, but we will use equilibrium BC for setup lid velocity and bounce back for the walls.

